

To get the most out of this project, follow this simple workflow to practice and test your skills.

- **Step 1: Read the Problem.** Open the main Problem PDF first and read through the questions so you know exactly what you are trying to solve.
- **Step 2: Review the Notes.** Read through the explanations in this document to understand the concepts and the logic behind the code.
- **Step 3: Try It Yourself.** Open your IDE and attempt to write the Python code to answer the questions on your own.
- **Step 4: Check Your Work.** Inside this project folder, I have included the actual **CSV dataset file** so you can load the data directly. I have also included my complete **Python (.py) file**. If you ever get stuck, or once you finish and want to compare your logic, just open my Python file to see the exact code I used to solve the problem

Project & Note Disclaimer.....

Please keep a few quick things in mind as you read through these notes and projects:

- **Course Origins:** I created these notes while working through the Pierian Data Machine Learning course (2020). The specific project questions and structural exercises are sourced directly from that course.
- **Modernized Code:** Because technology moves fast, some of the Python syntax and library commands in the original 2020 course are now outdated or deprecated. I researched, rewrote, and updated the code myself. The solutions in these PDFs strictly use modern, current syntax so your code won't crash!
- **AI Collaboration:** I used AI to help write and format these notes. Because typing out every detailed explanation would take forever, my workflow involved explaining my logic, code, and thoughts to the AI via voice messages. I supervised and directed the content, while the AI acted as my scribe to make it highly readable and structured for you.
- **Data Sources:** The datasets used in these projects are publicly sourced from Kaggle. For example, the 911 Emergency dataset is the "Montgomery County, PA 911" dataset uploaded by Mike Chirico ([mchirico/montcoalert](https://www.kaggle.com/mchirico/montcoalert)). Other project files, like the Advertising data, are standard open-source datasets provided for educational use.